GTPerf

User manual

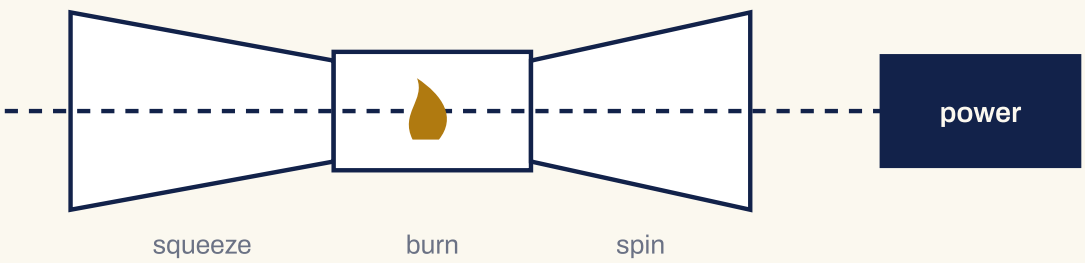
A plain-English guide to GTPerf, an open design and research tool for gas turbine performance. No engineering background needed.

Edition 2: design mode with calculated compressor and turbine maps

Tool: **tanishq-charde.github.io/gtperf-app**

Version 1.3 · October 2026

Author: Tanishq Charde



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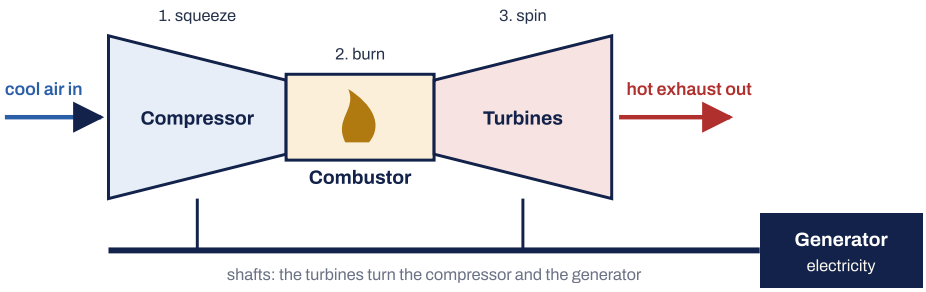
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1. What a gas turbine is

A gas turbine is the engine inside many power stations and every jet plane. It does three things, continuously: it **squeezes** air, **burns** fuel in it, and lets the hot gas **spin** a turbine.



THINK OF IT LIKE THIS

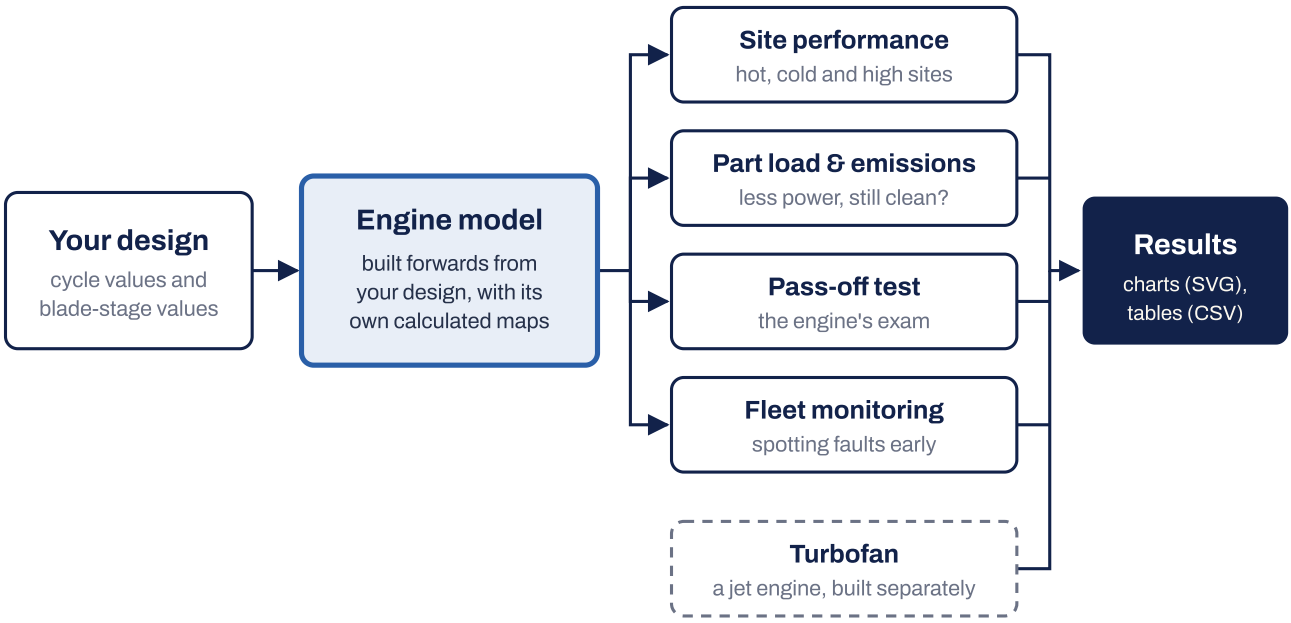
Blow into a pinwheel and it spins. A gas turbine does the same, but the "wind" is burning gas at over 1,000 °C, so it spins hard enough to drive a generator.

Performance answers two questions: how much power does the engine make (in megawatts, MW), and how much fuel does it burn to make it? Less fuel for the same power is better.

THE BIG PICTURE

2. What GTPerf does

GTPerf is a **design and research tool**. You set an engine's design: how much air it takes in, how hard it squeezes it, how hot it runs and how its blades are laid out. GTPerf builds that engine, works out how each part behaves and shows how every choice affects power, fuel use, emissions and safety margins, so you can explore options and understand the trade-offs.



Read left to right. You design the engine once on Engine design; every study on the right then uses it. The jet engine (Turbofan) is designed from its own values on its own page.

THINK OF IT LIKE THIS

A flight simulator lets a pilot try a storm without risking a real plane. GTPerf lets you try a new blade design, a hotter turbine, a heatwave or a worn engine without building anything, and see exactly why the result changes. The models are open Python code, so they can be extended for your own research.

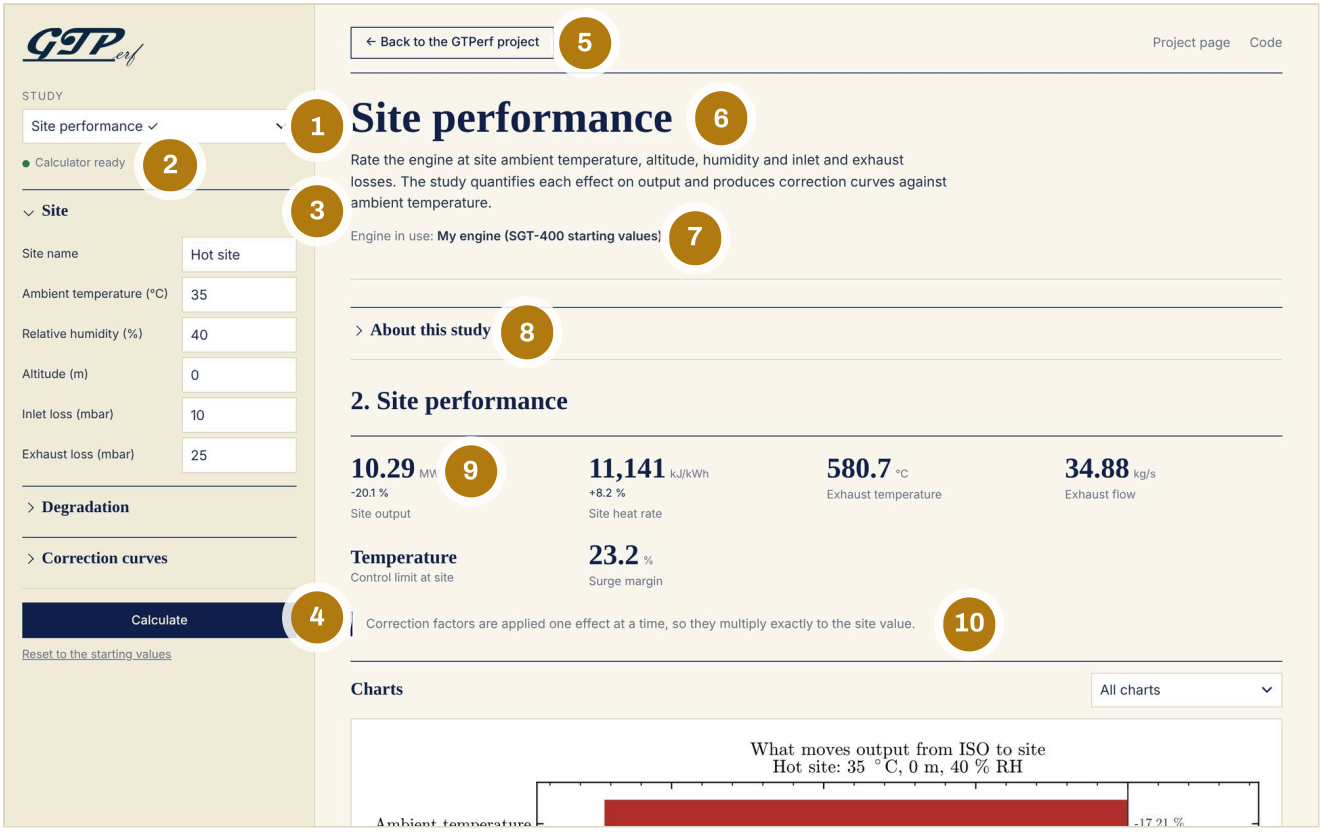
Good to know

- The inputs start at values that reproduce a real engine, the **Siemens Energy SGT-400** (12.9 MW), built in Lincoln, UK. Change any of them and it becomes your own design.
- **No input has a limit.** You can type any number; the tool tells you in plain words if a value cannot describe a real engine.
- Everything is calculated **inside your web browser**. Your results stay while the page is open; reloading clears them.

FINDING YOUR WAY

3. The screen, part by part

Every page has the same layout: inputs in the sidebar on the left, results on the right. Learn it once and you can use every study.



1. **Study drop-down.** The main menu. Pick a study to open it; ✓ means it has results, ⚠ means it needs recalculating.
2. **Calculator status.** Green "Calculator ready" once the tool has loaded.
3. **Input groups.** The values you can change. Click a group's name to open or close it. Changed values are highlighted.
4. **Calculate button.** Runs the study with the values above it. "Reset to the starting values" is underneath.
5. **Back and links.** Back to the GTPerf project page, and the project's code.
6. **Title and description.** Which study you are in and what it answers.
7. **Engine in use.** Shown while you use the starting SGT-400 values; it disappears once you calculate your own design.
8. **About this study.** Click to read how the study works and what it is based on.
9. **Key figures.** The headline answers in large numbers with their units.
10. **Notes, charts and tables.** Scroll down for the detail (section 7).

> **TIP**
>
> Changing a value does nothing on its own. The tool recalculates only when you press the button under the inputs (**Calculate**, **Run the test** or **Simulate and monitor**).

HANDS ON

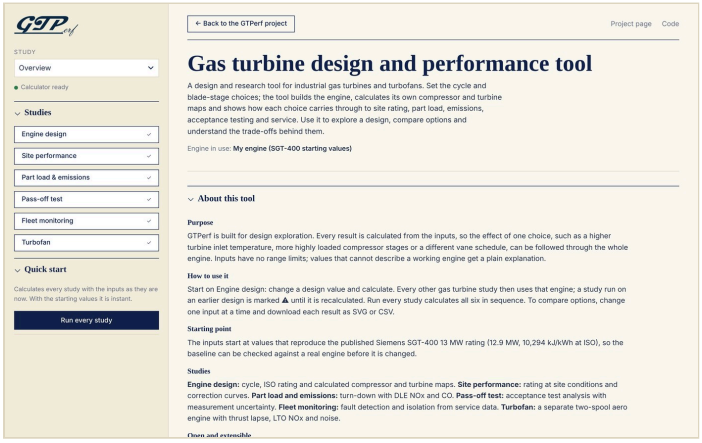
4. Your first five minutes

The quickest way to learn the tool is to let it calculate everything once, then look around.



1. **Open the tool** at **tanishq-charde.github.io/gtperf-app**, or press **Try the Tool** on the GTPerf project page, in Chrome, Edge, Firefox or Safari. The page needs an internet connection the first time, to download the Python calculator into the browser.
2. **Wait for "Calculator ready"** under the Study drop-down. The first load takes up to a minute; later visits are faster. The starting results are already on the page, so you can read them while it loads.
3. **Press "Run every study"** on the Overview page, under *Quick start*. With the starting values it is instant; with your own values it takes about a minute.
4. **Pick a study** with the **Study** drop-down, or the study buttons in the sidebar.
5. **Read and download.** Start with the key figures, then the charts and their captions, then the tables. Section 7 explains each part.

Then try your own numbers



The Overview page after "Run every study": every study in the sidebar has a ✓.

TIP

Change one value at a time, press the button and see what moves. For example, raise the ambient temperature on Site performance from 15 to 35 °C and watch the power fall by about a fifth.

DESIGN MODE

5. Designing your engine

On **Engine design** you set the engine's values in nine groups and press **Calculate**. The tool builds the engine forwards from them: nothing is fitted or looked up. Every other gas turbine study then uses that engine.

Input group	Starting values	What it controls
Air and compression	39.22 kg/s, PR 17.1, η_p 0.89	How much air the engine swallows, how hard the compressor squeezes it (pressure ratio) and how efficiently. Air flow sets the engine's size.
Combustion and turbines	TIT 1280 °C, η 0.8685, 4 % loss	How hot the gas is entering the turbine (TIT), how well the turbines extract work, and the pressure lost in the combustor. TIT is the strongest lever on output.
Turbine cooling air	7 %, 5 %, 3 %, 0.6	Air taken from the compressor to cool the hot turbine parts. Cooling protects the metal but costs output and efficiency.
Shaft and generator	0.99, 0.98, 1.03, 20 MVA, 0.8	Shaft and generator losses, the gas generator speed limit and the generator's rating. These limits cap output on cold days.
Compressor stages	11 stages, ψ 0.40, φ 0.50	The blade design that the compressor map is calculated from: number of stages, stage loading ψ (work per stage), flow coefficient φ, reaction, solidity and the diffusion factor at which a blade row stalls.
Variable vanes	3 rows, −35°, 0.95 / 0.85	Adjustable inlet vanes that close at low speed to keep the front stages from stalling: how many rows, how far they close, and between which speeds.
HP turbine stages	2 stages, ψ 1.5, φ 0.6	The blade design of the turbine that drives the compressor, from which its map is calculated.
Power turbine stages	2 stages, ψ 1.3, φ 0.6	The blade design of the free turbine that drives the generator.

THINK OF IT LIKE THIS

Think of a staircase. The pressure ratio is the total height to climb; the number of stages is the number of steps; stage loading is how tall each step is. Taller steps are quicker but easier to trip on: the compressor gets closer to stalling.

TIP

What you get back: the ISO rating (output, heat rate, efficiency, exhaust conditions), the surge margin, the gas path temperatures and pressures, and the calculated compressor, HP turbine and power turbine maps. When you change the engine, the other studies show ⚠ until you run them again.

DESIGN MODE

6. How the maps are made, and how to read them

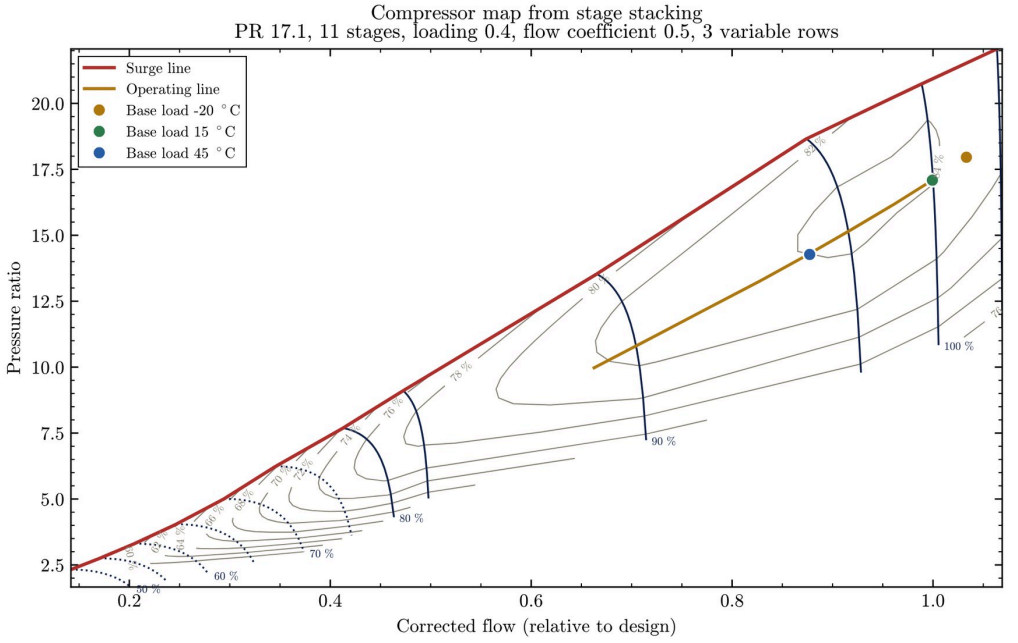
A **map** is a chart of how a compressor or turbine behaves at every speed and flow. GTPerf does not stretch a stored map to fit; it **calculates** the map from your blade-stage values, one stage at a time (stage stacking).

How a map is calculated

1. The design work is shared out over the stages.
2. Stage loading and flow coefficient fix each stage's blade speed and the angles of the flow on and off the blades.
3. The blade angles and flow areas are then **frozen**, as they are in metal.
4. At every other speed and flow, each stage is worked out in turn. When the flow meets a blade at the wrong angle, extra losses appear.
5. A compressor blade row **stalls** when its diffusion factor reaches the limit you set. A turbine row **chokes** when the gas through it reaches the speed of sound.

What to look for

- **Speed lines** (navy): one per shaft speed, labelled in % of design.
- **Surge line** (red): the edge of stable operation. Beyond it the flow breaks down and can reverse.
- **Operating line** (amber): where the engine actually runs as load changes.
- **Surge margin**: the gap between the operating point and the surge line. About 15 to 25 % is typical; the starting engine has 21 %.
- **Dotted lines**: a blade row past its stall limit, a warning region.



Compressor map

Download SVG

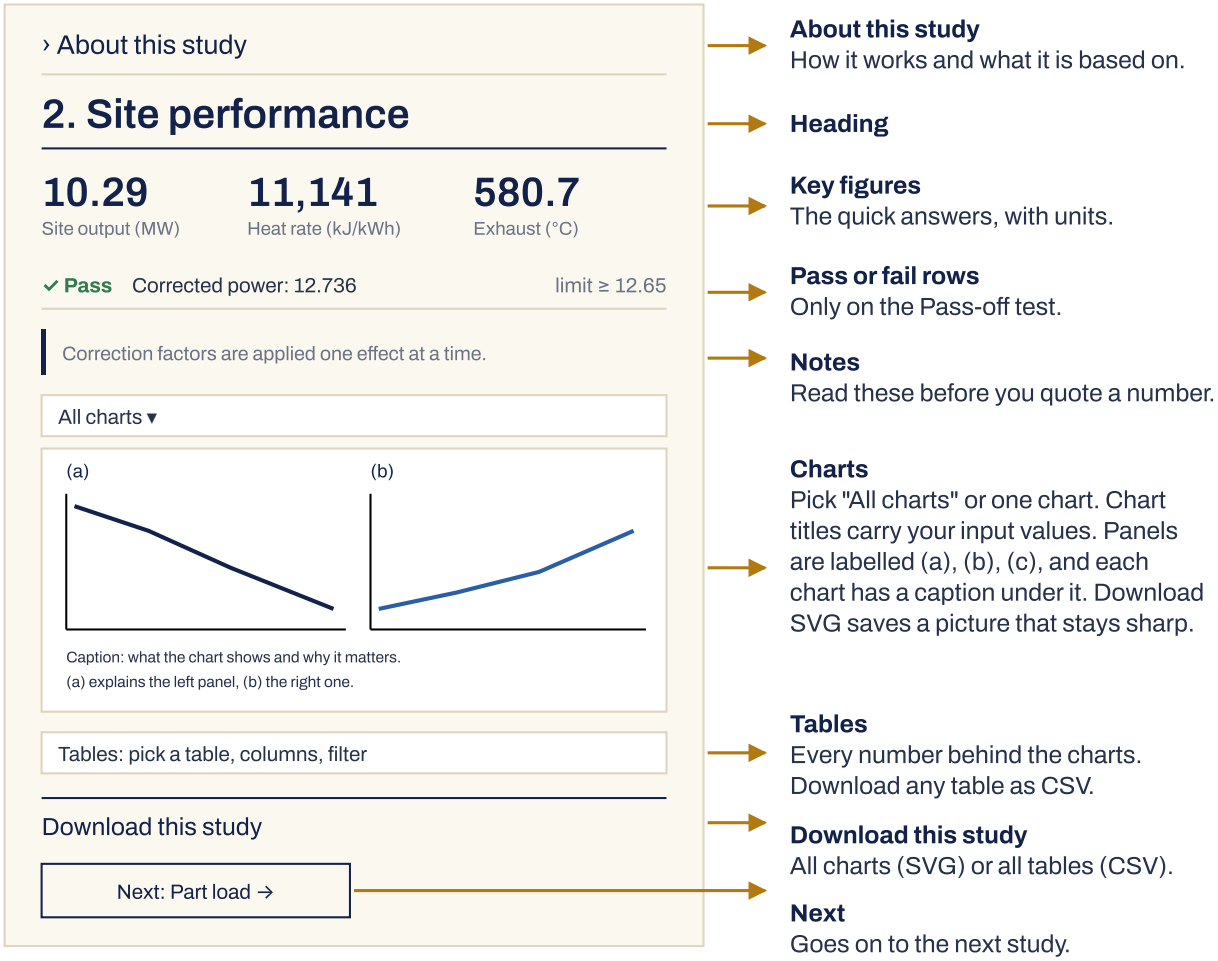
Pressure ratio against corrected flow for each speed line, with efficiency contours. The red surge line is the stability boundary; dotted lines have a blade row past its diffusion limit. The operating line and the base-load points at −20, 15 and 45 °C show where the engine runs. The vertical gap to the surge line is the surge margin.

The compressor map of the starting engine, as the tool draws it, with its caption. Base-load points from −20 to 45 °C show how the running point moves with the weather. Turbine maps show flow capacity in panel (a) and efficiency in panel (b) against pressure ratio.

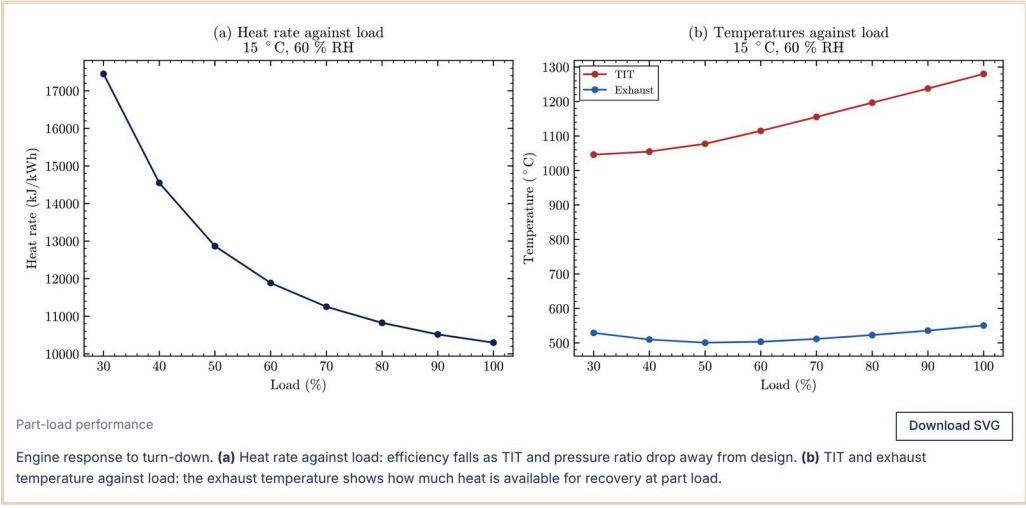
UNDERSTANDING THE ANSWERS

7. How to read a results page

Every study lays out its answers in the same order, from the quick answer at the top to the full detail at the bottom.



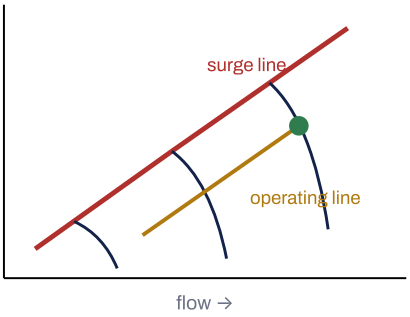
A simplified drawing of a results page.



A real two-panel chart: panels (a) and (b), and the caption that explains each.

8. The six studies

Each study answers one question. Design the engine first; the other gas turbine studies then use it. The Turbofan is independent.



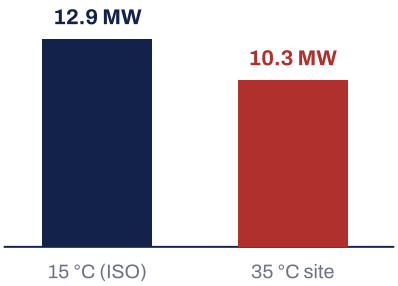
Engine design

"What does my design produce, and is it stable?"

You set the cycle (air flow, pressure ratio, TIT, efficiencies, cooling, limits) and the compressor, vane and turbine stage design.

You press Calculate.

You get the ISO rating (12.89 MW, 10,299 kJ/kWh, 35.0 % for the starting values), the surge margin (21.3 %), the gas path, and the calculated compressor and turbine maps.



Site performance

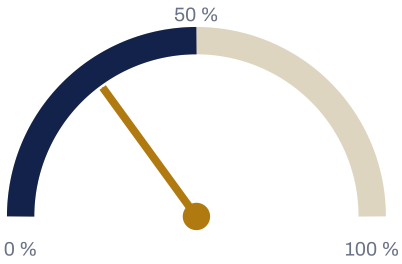
"How much power will the engine make at my site?"

You set the site: temperature, humidity, altitude and the pressure losses in the inlet and exhaust ducts. You can also add wear (degradation) and choose the range of the correction curves.

You press Calculate.

You get output, heat rate and exhaust conditions at the site, what each effect costs, which limit controls the engine, and correction curves from cold to hot days.

> **THINK OF IT LIKE THIS**
>
> Hot air is thinner, so the engine swallows less of it each second, and less air means less power. On a 35 °C day the starting engine loses about a fifth of its rating.



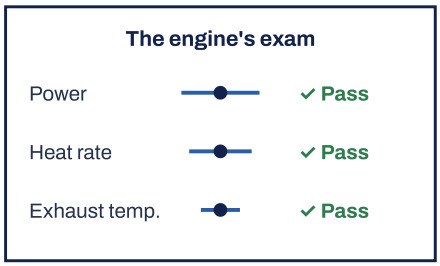
Part load and emissions

"What happens below full power, and is the exhaust still clean enough?"

You set the weather, how far to turn down and in what steps, DLE air staging on or off, and the NOx and CO limits.

You press Calculate.

You get heat rate, temperatures, NOx and CO at each load, and the lowest load inside both limits (30 % for the starting engine).



blue bar = measurement uncertainty

Pass-off test

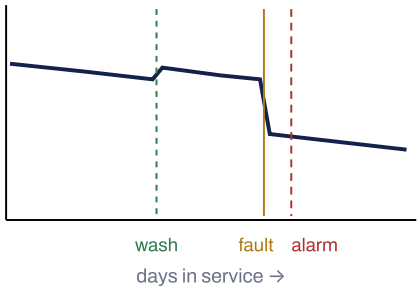
"Would this engine pass its acceptance test, and how sure are we?"

You set the test-day weather, the number of test points, which sensors go wrong, the engine's hidden health, the guaranteed values and the number of Monte Carlo trials.

You press Run the test.

You get a pass or fail for each guarantee, each corrected result with its ± uncertainty (12.736 ± 0.083 MW to start with), the faulty sensors that were rejected, and a check against the hidden truth.

> **THINK OF IT LIKE THIS**
>
> Like an exam with a strict examiner: first check the measuring tools and throw out the broken ones, then correct for the weather on the day, then mark the engine against what was promised. It follows the ASME PTC 22 and ISO 2314 test standards.



Fleet monitoring

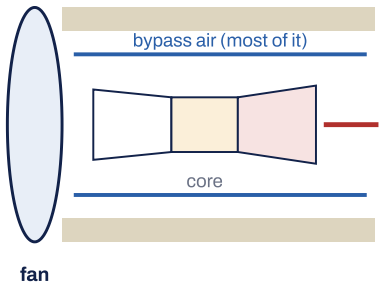
"If engines go wrong in service, do we notice, how quickly, and do we blame the right part?"

You set the number of engines, days in service, days between data snapshots, base load or cycling, the faults to inject, and which engine to plot.

You press Simulate and monitor.

You get how many faults were found (5 of 5 to start with), how many were blamed on the right part (3 of 5), false alarms, and trend charts for the chosen engine.

> **THINK OF IT LIKE THIS**
>
> Like a fitness tracker noticing your resting heart rate creeping up: slow changes are easy to miss by eye, so the tool watches the trends for you.



Turbofan

"For a jet engine of my design: cruise fuel use, take-off thrust, airport NOx and noise?"

You set the cycle (bypass ratio, pressure ratios, turbine temperatures), the cruise thrust, altitude and speed, the combustor's NOx level, and the fan, booster, compressor and turbine stage design.

You press Calculate.

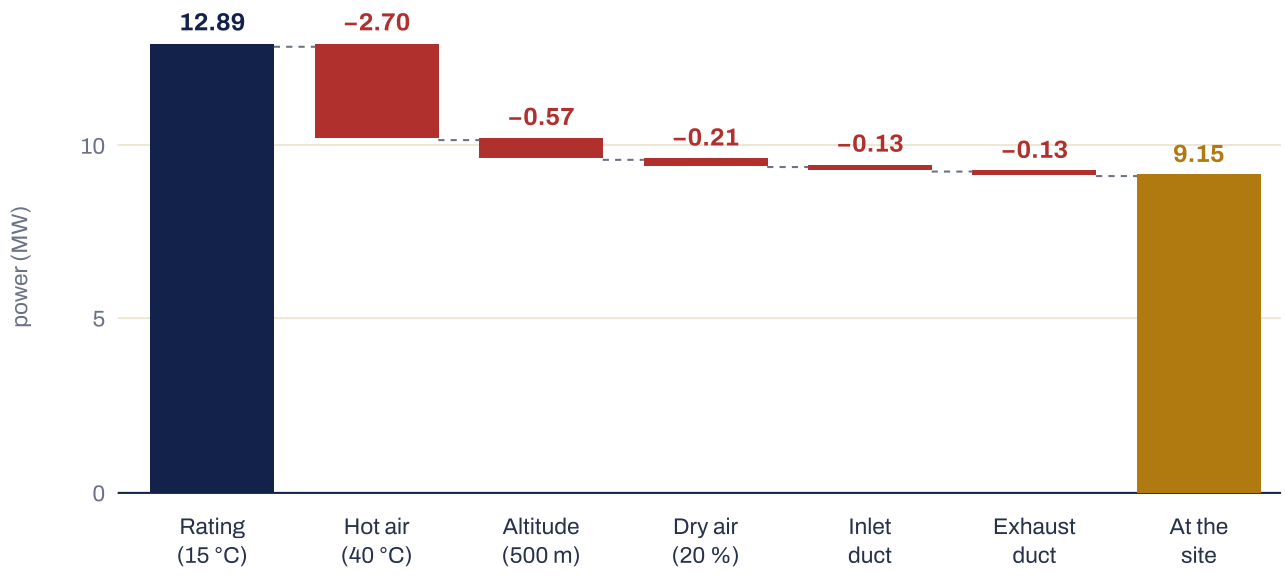
You get cruise fuel use per unit thrust (TSFC, 1.527×10^{-5} kg/(N s) to start with), thrust at take-off and altitude, NOx at the four ICAO landing and take-off modes against the CAEP limits, noise sources, and the calculated fan, HPC, HPT and LPT maps.

TRY IT YOURSELF

9. Worked example: a hot desert site

The question: the starting engine is going to a desert site at 40 °C, 500 m above sea level, with dry air (20 % humidity). How much power will it make? **The answer: 9.15 MW, which is 29 % below its 12.89 MW rating.**

1. Pick **Site performance** in the Study drop-down.
2. Open the **Site** group. Type: site name *Desert site*, ambient temperature *40*, relative humidity *20*, altitude *500*, inlet loss *10*, exhaust loss *25*.
3. Leave **Degradation** at zero: this is a new engine.
4. Press **Calculate** and read the key figures: 9.15 MW, heat rate 11,331 kJ/kWh, exhaust 586.4 °C, control limit "temperature", surge margin 23.4 %.
5. Look at the **Correction factors** chart to see where the power went. It is redrawn below as a staircase.



Each red step is power lost to one effect. Heat alone takes away 2.70 MW, more than all the others together.

THINK OF IT LIKE THIS

Hot air and high places both mean thinner air. It is like breathing on a mountain top: the engine has less air to work with, so it makes less power.

What "Control limit: temperature" means. The engine is held back by how hot its turbine may run (TIT), not by how fast it can spin. On cold days the gas generator speed limit takes over instead; the correction curves show where.

TRY IT YOURSELF

10. Worked example: changing the design

The question: what do a hotter turbine and a more highly loaded compressor do to the engine? Two changes on **Engine design**, one at a time, each followed by **Calculate**. Press **Reset to the starting values** between them.

Change A: run the turbine hotter

Open **Combustion and turbines** and raise the turbine entry temperature from *1280* to *1350* °C.

Result	Starting engine	TIT 1350 °C	Change
ISO output	12.89 MW	14.36 MW	+11.4 %
Heat rate	10,299 kJ/kWh	10,120 kJ/kWh	−1.7 % (better)
Thermal efficiency	35.0 %	35.6 %	+0.6 points
Exhaust temperature	550.5 °C	590.2 °C	+39.7 K

A 70 K hotter turbine gives over a tenth more power from the same engine size. That is why engine makers invest so much in turbine materials and cooling. The price is hotter metal, a shorter life, and more NOx (check it on Part load and emissions).

Change B: load the compressor stages more heavily

Open **Compressor stages** and raise the stage loading ψ from *0.40* to *0.50*. The output stays 12.89 MW, because the cycle has not changed. The compressor map has: the **surge margin falls from 21.3 % to 5.6 %**.

THINK OF IT LIKE THIS

Back to the staircase: the same climb, with taller steps. Each blade row now has to slow the air down more sharply, so it gets close to stalling. A 5.6 % margin would not be accepted in service, where 15 % or more is usual.

TIP

This is what design mode is for. Most changes on the map pages leave the ISO rating the same but change how close the engine runs to its limits. Compare the compressor map before and after: the red surge line moves down towards the operating point.

HELP

11. When something looks wrong

Most surprises come from four things: the first load is slow, numbers update only when you press the button, some values cannot describe a real engine, and some data is made up on purpose.

What you see	What to do
"Calculator loading" does not finish	Wait up to a minute; Python is downloading into your browser. Check you are online. Next time it is faster.
I changed a number and nothing happened	Press the button under the input groups. Values count only when you press it.
A study shows ⚠	You changed the engine after that study ran. Open the study and press its button again.
"Engine in use" has disappeared	That is expected: you are using your own design. Reset to the starting values on Engine design and calculate to bring it back.
"The calculation did not work for these values"	Read the line under it. A plain reason (for example an efficiency above 1, or vanes still part closed at design speed) tells you which value to fix. "Could not find a matched running point" means the values are too far from a workable engine: move the ones you changed back toward the starting values.
"The compressor has no stable operating range"	The compressor stage design cannot run at its own design speed. Lower the stage loading, change the flow coefficient, or add stages.
A thrust lapse curve stops early	That spool has reached its highest calculated speed line. A real engine would be held there by its speed limits.
The page seems busy for a long time	Fleet monitoring takes the longest, especially with cycling dispatch (a few minutes in the browser). Wait for the spinner to stop.
All my results disappeared	Reloading the page clears everything. Download the charts and tables you need first.

What the numbers are, and what they are not

- The starting values reproduce the **published** SGT-400 rating. Details inside the model, such as blade-stage values and cooling flows, are sensible engineering choices, not the maker's data.
- The Pass-off test and Fleet data are **synthetic on purpose**, with hidden faults, so the tool can check its own answers against the truth.
- Turbofan noise gives source levels for comparing designs, not an official certification noise level.
- GTPerf is a **learning and portfolio tool**. Use it to understand and compare engines, not to certify or sell one.

HELP

12. Words you will see

Word	What it means
MW (megawatt)	A unit of power. 1 MW supplies roughly 1,000 homes. The starting engine makes about 13 MW.
ISO conditions	The standard day for comparing engines: 15 °C, sea level, 60 % humidity.
Heat rate (kJ/kWh)	Fuel energy needed for one unit of electricity. Lower is better.
Thermal efficiency	The share of the fuel's energy that becomes electricity, about 35 % here. Higher is better.
Pressure ratio (PR)	How hard the compressor squeezes the air: 17 means 17 times the outside pressure.
TIT / TET	Turbine inlet (entry) temperature: how hot the gas is entering the turbine. Hotter gives more power but wears the metal faster.
Stage	One row of moving blades plus one row of fixed vanes. A compressor has many; a turbine has a few.
Stage loading ψ	How much work each stage does compared with its blade speed squared. Higher means fewer, harder-working stages.
Flow coefficient φ	How fast the air moves through the stage compared with the blade speed.
Reaction	How the pressure change is split between the moving and fixed blades of a stage.
Diffusion factor	How sharply a blade row slows the air. Past a limit (about 0.6) the flow separates and the row stalls.
Variable vanes (VIGV)	Adjustable vanes at the compressor front that close at low speed to keep it stable.
Map, speed line	A chart of a compressor's or turbine's behaviour; each speed line shows one shaft speed.
Surge line, surge margin	The compressor's stability edge, and the safety gap between it and where the engine runs.
Choke	When the gas through a blade row reaches the speed of sound, so no more flow can pass.
Part load	Running below full power, for example at 50 %.
NOx, CO, DLE	Two pollutants from burning fuel, and the "dry low emissions" combustor that keeps them low.
Pass-off test, guarantee	The acceptance test a new engine must pass, and the values the maker promises in it.
Uncertainty (± U95)	How sure we are: the true value lies inside the ± band with 95 % confidence.
Monte Carlo trials	Repeating the analysis many times with random small errors to check the uncertainty.
Gas path analysis	Working out which part of the engine changed from the pattern of measured changes.
Bypass ratio, TSFC	In a jet engine: air going around the core compared with through it; and fuel used per unit of thrust.
LTO cycle, EI, Dp/Foo	The ICAO landing and take-off routine; grams of NOx per kg of fuel; and grams of NOx per kN of thrust over the cycle.
CAEP/6, CAEP/8	International NOx limits for new aircraft engines. CAEP/8 is the stricter one.